

Lecture 1.8 – Lists and Data Frames

Specific Learning Objectives:

1.1.10 – Create vectors, arrays, matrices, lists, and data frames.

1.1.11 – Understand vectors and vectorized calculations.

1.1.12 – Learn how to index vectors, arrays, matrices, lists, and data frames.

Check Your Understanding

Which class of object would you use if you needed:

- a) Members of different sizes
- b) Members of different classes
- c) Both a and b

List	Data frame
	
	
	

Check Your Understanding

Create a list in which each member contains one of each data types you've learned so far in the course!

Check Your Understanding

In the `ToothGrowth` data set, how can you print out all the measured tooth lengths from their study?

How can you find the mean and standard deviation of these lengths?

Check Your Understanding

In the `ToothGrowth` data set, how can you print out all the measured tooth lengths from their study that were only given a dose of 1.0?

How can you find the mean and standard deviation of these lengths?

Can you find the mean of the tooth lengths for animals given an OJ dose of 1.0?

In-class Problem Review

5. Predict the output of the following operation:

```
my_array[-6, , 1] * diag(x = 5)
```

Davies Ch 3.3.2

```
> diag(x=5)
      [,1] [,2] [,3] [,4] [,5]
[1,]    1    0    0    0    0
[2,]    0    1    0    0    0
[3,]    0    0    1    0    0
[4,]    0    0    0    1    0
[5,]    0    0    0    0    1
```

Answer:

```
      [,1] [,2] [,3] [,4] [,5]
[1,]    0    0    0    0    0
[2,]    0   21    0    0    0
[3,]    0    0   42    0    0
[4,]    0    0    0   63    0
[5,]    0    0    0    0   84
```

, , 1

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	0	18	36	54	72
[2,]	3	21	39	57	75
[3,]	6	24	42	60	78
[4,]	9	27	45	63	81
[5,]	12	30	48	66	84
[6,]	15	33	51	69	87

, , 2

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	90	108	126	144	162
[2,]	93	111	129	147	165
[3,]	96	114	132	150	168
[4,]	99	117	135	153	171
[5,]	102	120	138	156	174
[6,]	105	123	141	159	177

, , 3

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	180	198	216	234	252
[2,]	183	201	219	237	255
[3,]	186	204	222	240	258
[4,]	189	207	225	243	261
[5,]	192	210	228	246	264
[6,]	195	213	231	249	267

In-class Exercises

1. Questions about past worksheets/assignments?
2. Worksheet!!

Action Items

- 1. Complete Assignments 1.9 and 1.10.**
- 2. Read Davies Ch. 6 for next time.**